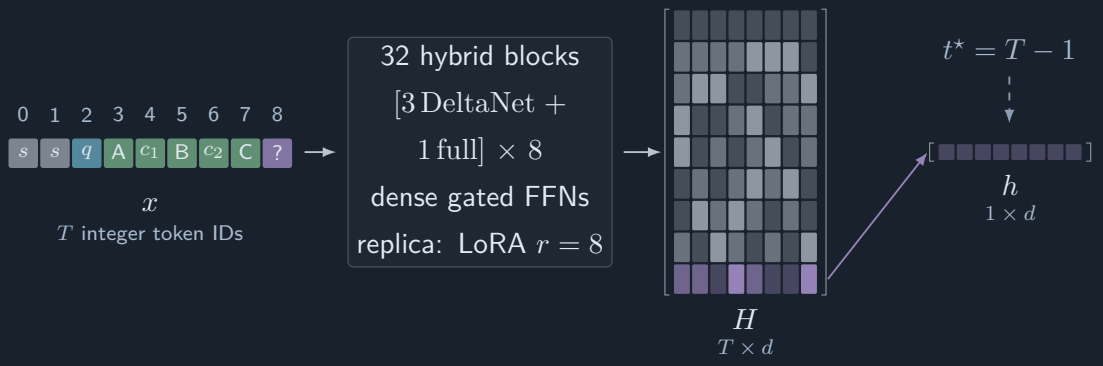
$$h = H_{t^\star,:}, \quad z = W_{\mathrm{lm}} h^{\mathsf{T}}, \quad s_k = z_{\ell_k}, \quad p_k = \mathrm{softmax}_k(s/\tau)$$

1. State, question and answer choices

2. Qwen3.5-4B

3. Last prompt position

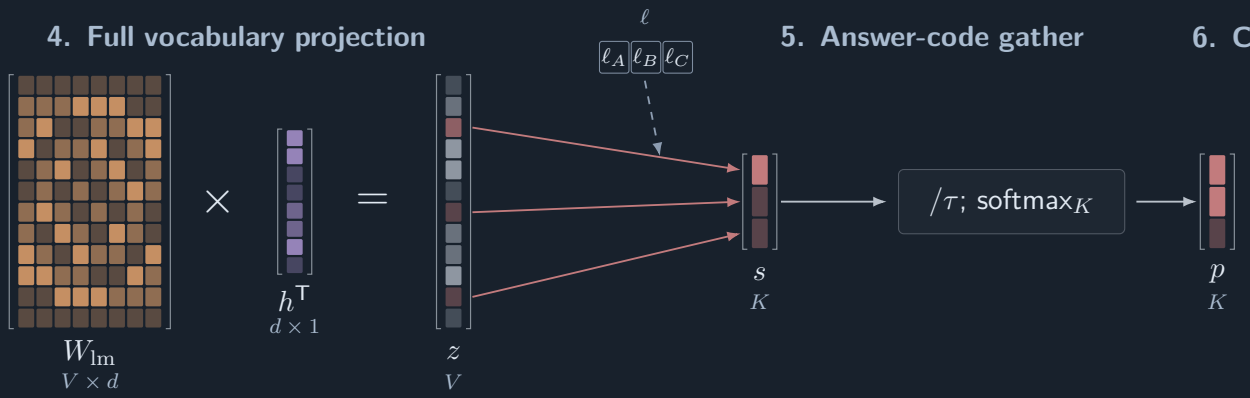


$d = 2560$, $V = 248320$
existing LM vocabulary head
no added scalar MLP
text branch; no rationale decoding

4. Full vocabulary projection

5. Answer-code gather

6. Candidate probabilities



ℓ_k: token ID of letter k
not a hidden-state position
W_{lm} tied to embeddings

Axes Single question; T: prompt length, $d = 2560$, $V = 248320$, K: valid choices. Tile counts illustrate axes. h is explicitly transposed before the GEMV.

Objects The replica computes all V logits, then gathers integer code IDs ℓ; no scalar head or selected-weight-only shortcut is implemented on this path. Softmax runs only over valid choices.

Mechanism Replica training: input prompt + gold letter; two full-vocabulary CE targets (letter, EOS). Upstream saved evaluation uses temperature-0 constrained generation; the replica reads logits directly. Accuracy reproduction does not recover historical weights/training.